

Assignment 1 – 1st-order modulator

- A. Design a 1st-order modulator with a sinc¹ decimation filter. What filter length is necessary to robustly ensure **8-bit** resolution over the modulator's entire input range? Show this by plotting the modulator's DC transfer function and INL over the input range:
 $-1 < x < 1$ **and** $-0.01 < x < 0.01$ [3 points]
- B. Repeat exercise A for a sinc² FIR filter. [2 points]
- C. Use a 1024-tap sinc¹ decimation filter, what is the minimum DC gain of the accumulator to achieve **9bit** resolution? [3 points]
- D. Repeat exercise C with a sinc² FIR filter. [2 points]
- **Submit an m-file that demonstrates this!**
Include comments indicating conclusions of minimum filter lengths and minimum accumulator gain.
- Assignment is due by **10:30am May 7th**
- Instruction about how to submit assignment are on Brightspace